

Machine Learning

Report: Anomaly Detection in Web Authentication Systems

Group Members

Waqas Maqsood	Reg# 24109121
Shuaib Ahmed Sheikh	Reg# 24109118
Abdul Rehman	Reg# 24109126

Submitted To Sir: Dr.Mazhar Ali

Machine Learning-Based Anomaly Detection in Web Authentication Systems: A Comparative Study of Naive Bayes and Random Forest Classifiers

Waqas Maqsood
Department of Cybersecurity
Szabist University
Karachi, Pakistan
mscys24109121@szabist.pk

Shuaib Ahmed Shaikh
Department of Cybersecurity
Szabist University
Karachi, Pakistan
mscys24109118@szabist.pk

Abdul Rehman
Department of Cybersecurity
Szabist University
Karachi, Pakistan
mscys24109126@szabist.pk

Abstract—Web authentication systems are essential parts of the modern cybersecurity environment, but they are still susceptible to numerous threats, such as unauthorized access, misuse of credentials, and internal threats. The paper introduces a complex machine learning-based method of abnormalities detection in web authentication logs with the help of supervised learning algorithms. Two classification models, Naive Bayes and Random Forest, were created and tested on an artificial identified dataset of 10,000 authentication sessions with features such as the session time, the number of failed attempts, behavioral score, the type of device used and the geographical location. The results of our experiment show that the Random Forest classifier was able to perfectly classify data with 100% accuracy, precision, recall, and AUC-ROC of 1.00 and Naive Bayes classifier was able to classify data with 94% accuracy and AUC-ROC of 0.95. The research gives information about importance of features to be used in detecting anomalies and explains the practicality of implementing such systems in a production setting.

Keywords—anomaly detection, web authentication, machine learning, random forest, naive bayes, cybersecurity, intrusion detection

I. INTRODUCTION

The web authentication systems are the initial barrier in safeguarding digital assets and user information. Nevertheless, these systems are under a consistent risk of advanced attackers who use different strategies like brute force attacks, credential stuffing, and social engineering [1]. Conventional security measures that are based on the rule of thumb are often not adaptable to the changing attack strategies, and the introduction of intelligent security systems that have the capability to learn and recognize anomalous activities is necessary [2]. Authentication system anomaly detection refers to detecting patterns that are abnormal to the normal user behavior [3]. These irregularities could help to signify the security incidents, including the account compromise, unauthorized access attempts, or insider threats. Machine learning methods have become an effective means to do so, as they provide an opportunity to learn complex patterns automatically on the basis of past data, as well as to change according to new threat landscapes [4]. This study examines how machine learning algorithms can be used to identify anomalies on web authentication logs through supervised algorithms. In particular, we compare the behavior of Naive Bayes and Random Forest classifiers which are two opposite

approaches in terms of concepts of probabilistic and ensemble respectively. The purpose of our study is to offer some information about the effectiveness of these algorithms in the context of security applications in the real world.

A. Research Objectives

The main goals of the study are threefold: (1) to carry out and test the autonomous machine learning scheme to detect abnormal authentication, (2) discover crucial features of distinguishing between normal and abnormal authentication events, and (3) compare the performance features of Naive Bayes and Random Forest classifiers in this area.

II. RELATED WORK

These methods of machine learning have heartily been used to detect what is out of place in cybersecurity [3], [6]. The first strategies were based on signature-based detection systems which works well in familiar threats situations, but does not work in new attack cases. The shift towards machine learning-driven and behavioral analysis has shown promising results as far as detecting previously unknown threats are concerned. The ability to work with high-dimensional data, and the possibility to define non-linear interactions have been demonstrated to render ensemble techniques, namely the Random Forests, more effective in a vast number of security tasks [1], [8]. Other probabilistic methods, including naive Bayes are simpler to compute and to understand, and are therefore attractive in resource-constrained environments [5].

III. METHODOLOGY

A. Dataset Description

We used artificially generated data in form of 10,000 web authentication logs. The synthetic methodology has been selected because it provides the solution to the privacy issues related to the actual authentication data and still allows us to model various scenarios. The data set has nine attributes of different aspects of authentication sessions. These attributes include categorical and numerical ones: User ID, Timestamp, Login Status (Success/Fail), IP Address, Device Type, Location, Session Duration (seconds) and Failed Attempts (count) and Behavioral Score (0- 100). Anomaly is the target variable, which is binary (0: Normal, 1: Anomalous) and has the proportion of around 18.15 in the dataset, which is a realistic scenario of a class imbalance.

B. Feature Engineering

The idea of feature engineering was instrumental in the model performance. Temporal characteristics were identified based on the timestamps such as hour of day and day of week to identify temporal trends in authentication behavior. One-hot encoding was used to provide all the category information of categorical variables into the encoded form without any ordinal relationship between them [4]. All the numerical features were standardized with the help of Standard Scaler so that all features impacted the model training in equal proportions [4]. This preprocessing step is especially relevant to those algorithms that rely on the independence of features and certain distributional properties such as Naive Bayes [5].

C. Anomaly Labeling Criteria

The identification of the anomalous sessions was made on several criteria which represented the security issues in the real world: (1) long session durations that reflected possibility of the session hijacking, (2) repeated and unsuccessful logins indicating the presence of brute force attacks, (3) abnormally low or abnormally high behavioral scores, which indicate automated or suspicious activity, (4) logins carried out via rare IP addresses, and (5) logins to unusual geographical locations.

D. Model Selection and Training

To compare two classification algorithms, it was identified that there are two different algorithms with unique features and common use in the sphere of security. Naive Bayes Classifier: Naive Bayes classifier is based on the independence of the features and it assumes that each feature is distributed as a Gaussian probability density when the features are modelled [5]. Naive Bayes can be useful in real-time applications and has a tendency to be effective in practice despite its simplifying assumptions, thus making it computationally efficient [2]. Random Forest Classifier: The random forest is a type of ensemble learning model that builds a series of decision trees in a training process, and the mode of the prediction made by all the trees is the output [1], [8]. To balance the performance and the cost of calculation we set up the model with 100 estimators. Random Forests is natural in dealing with interactions among features and resistant to outliers and noise. Stratified sampling was used to divide the dataset into training (80% and testing (20% sets) to ensure a balanced distribution of classes [4]. Fitting all preprocessing transformations on the training set and applying them to the test set were done to avoid data leakage.

IV. EXPERIMENTAL RESULTS

A. Exploratory Data Analysis

The first analysis showed some specific trends between normal and abnormal sessions. The distributions of session duration were abnormal at the two ends: very short sessions with the implication of automated scanning and very long ones with the implication of unauthorized access. More consistent and moderate durations were observed in normal sessions. There was evident discriminative power in failed attempts to log in, normal sessions exhibited zero failures predominantly whereas anomalous sessions exhibited a distribution over a range of failure counts. Behavioral scores showed that normal sessions are clustering higher values and abnormalities have greater ranges with bias towards lower scores. Correlation analysis resulted in the moderate positive correlation (0.45) between failed attempts and anomaly labels, and negative correlation (-0.52) between behavioral

scores and anomalies, which validates the significance of these characteristics in classification.

B. Random Forest Performance

Random Forest classifier was remarkable in all metrics. Table I shows the outcomes of the detailed classification.

TABLE I: RANDOM FOREST CLASSIFICATION RESULTS

Class	Precision	Recall	F1-Score	Support
Normal (0)	1.00	1.00	1.00	1653
Anomaly (1)	1.00	1.00	1.00	347
Accuracy	1.00			2000

The confusion matrix revealed perfect classification with 1653 true negatives, 347 true positives, and zero false positives or false negatives. The ROC curve showed ideal performance with an AUC score of 1.00, indicating perfect discrimination between classes at all threshold values [7].

C. Naive Bayes Performance

Naive Bayes classifier showed good but not perfect performance. The measures of classification are summarized in Table II.

TABLE II: NAIVE BAYES CLASSIFICATION RESULTS

Class	Precision	Recall	F1-Score	Support
Normal (0)	0.93	1.00	0.96	1653
Anomaly (1)	1.00	0.65	0.79	347
Accuracy	0.94			2000

The model recorded 94% overall accuracy and AUC-ROC of 0.95. It is worth noting that it had a 1.00 precision (perfect) and a lower recall (0.65), which left 121 false negatives. The confusion matrix indicated that all the 1653 normal cases were correctly identified with only 226 out of 347 anomalies identified.

D. Comparative Analysis

In Table III, the two models are shown in a direct comparison on key performance metrics.

TABLE III: MODEL PERFORMANCE COMPARISON

Metric	Random Forest	Naive Bayes
Accuracy	1.00	0.94
Precision (Anomaly)	1.00	1.00
Recall (Anomaly)	1.00	0.65
F1-Score (Anomaly)	1.00	0.79
AUC-ROC	1.00	0.95
False Negatives	0	121
False Positives	0	0

The excellent performance of the Random Forest is explained by the fact that this algorithm is capable of describing complex interactions of features and non-linear correlations [1], [8]. The ensemble method decreases variation and enhances generalization. Naive Bayes is efficient but it has the weakness of its independence assumption, which

might not be true with authentication features that have common correlations [5].

V. DISCUSSION

A. Implications for Security Systems

The optimal results of the Random Forest on the test set are indicative of a high possibility of adoption in production settings. Practitioners are however advised to be wary of overfitting especially when working with synthetic data versus real world data. The fact that the model has a zero false positives capability is especially useful, as it reduces the chances of preventing genuine users. The high accuracy of Naive Bays (1.00) of anomalies implies that when Naive Bays sensitizes an event as an anomaly, it is always right in our test set. Nevertheless, recall is 65% which means it detects 35% of true anomalies [7]. This trade-off can be tolerated in security situations in which false alarms are especially expensive, but it becomes dangerous in application areas where false negative attacks are critically fatal.

B. Feature Importance and Interpretability

Graphical analysis of feature contributions showed that the top features were: failed login, behavioral scores and duration of stay. This is consistent with the knowledge of security domain, where failure to happen many times can indicate attempts of attacks and unusual session features can be indicative of compromised accounts [6]. Random Forest offers built-in importance ranking of features (Gini importance), which is interpretable because it is an ensemble model anyway [8]. Naive Bayes, which, in theory, is more interpretable in terms of probability estimates, had the problem of correlated features, which restrict its interpretability in this problem context.

C. Computational Considerations

Naive Bayes training took a much shorter time than Random Forest (about 10 times shorter), so it would be desirable in situations where updating the model frequently was needed. Yet, the better accuracy of the Random Forest can make the increased calculation cost worthwhile in security sensitive situations where accuracy of detection should be the most important factor.

D. Limitations and Future Work

The first weakness of this study is the use of synthetic data, which might fail to reflect all the complexities of the authentication logs in the real world. Future work would need to substantiate these results on production data and resolve the privacy issue with such methods as the differentiated privacy or federated learning. The ideal work of Random Forest evokes the doubts on possible overfitting or data leakage. The experiments of cross-validation and testing on

completely different datasets would help to enhance faith in the ability of the model to generalize. Moreover, by exploring more aggressive algorithms like deep-learning models, unsupervised based anomaly detection, or hybrid schemes based on a combination of algorithms, the detection will be enhanced further. The aspect of real-time deployment such as model updating strategy and concept drift should be investigated.

VI. CONCLUSION

This study involved an all-inclusive assessment of machine learning methods in detecting abnormalities in web authentication systems. Our experiments proved that the supervised learning, especially Random Forest classifiers, can reach outstanding results in detecting the cases of anomalous authentication. The Random Forest model showed a 100% accuracy, precision, recall and an AUC-ROC of 1.00 which is perfect whereas Naive Bayes showed a good performance with an 94 percent accuracy and a 0.95 AUC-ROC. Recall to detect anomalies was the point of difference as Random Forest identified all the anomalies and Naive Bayes failed to identify 35 percent. In a practical deployment we can suggest Random Forest in highly secured settings where anomalies missing cannot be tolerated even though it will be more expensive computationally. Naive Bayes can still be used in cases where resources are limited or where false alarms are required to be kept to a minimum at the cost of detection coverage. Future research should aim at validation to real-world data, exploration of unsupervised and semi-supervised methods in situations with sparse labelled data as well as development of adaptive systems that can adapt to emerging threat patterns. The adoption of explainable AI methods would also make production security systems more trusted and adopted.

REFERENCES

- [1] L. Breiman, "Random Forests," *Machine Learning*, vol. 45, no. 1, pp. 5-32, 2001.
- [2] T. M. Mitchell, *Machine Learning*. New York: McGraw-Hill, 1997.
- [3] V. Chandola, A. Banerjee, and V. Kumar, "Anomaly detection: A survey," *ACM Computing Surveys*, vol. 41, no. 3, pp. 1-58, 2009.
- [4] F. Pedregosa et al., "Scikit-learn: Machine Learning in Python," *Journal of Machine Research*, vol. 12, pp. 2825-2830, 2011.
- [5] H. Zhang, "The Optimality of Naive Bayes," in *Proc. FLAIRS Conference*, 2004.
- [6] M. Ahmed, A. N. Mahmood, and J. Hu, "A survey of network anomaly detection, techniques," *Journal of Network and Computer Applications*, vol. 60, pp. 19-31, 2016.
- [7] T. Fawcett, "An introduction to ROC analysis," *Pattern Recognition Letters*, vol. 27, no. 8, pp. 861-874, 2006.
- [8] G. Louppe, "Understanding Random Forests: From Theory to Practice," *arXiv preprint arXiv:1407.7502*, 2014.